<u>Magnetic Diagnostics – Coordinates Of The Sensors</u>

Ted Strait and S. Munaretto – Last update: 5/5/20

This memo documents the names, positions and dimensions of the DIII-D magnetic sensors. There are two main sections:

I. Individual Sensors

II. Differential Pairs

The following sets of sensors are included. Major arrays for "3D magnetics" are indicated **in bold** (see Part II for additional details).

POLOIDAL FIELD PROBES

Bp – 322 degree poloidal array
Bp – 67 degree backup probes
Bp – 142 degree poloidal array
Bp – upper divertor baffles
Br – upper divertor baffles
Bp – R0 ELM array (127-137 deg.)
Bp – R0 ELM array (307-317 deg)
Bp – lower divertor toroidal array
Bp – R0 toroidal array
Bp – R+1 toroidal array
Bp – R-1 toroidal array
Bp – R+2 toroidal array
Bp – R-2 toroidal array
Bp – HFS vertical array (139 deg)
Bp – HFS vertical array (199 deg)
Bp – HFS toroidal array (Above midplane)
Bp – HFS toroidal array (Below midplane)

RADIAL FIELD LOOPS

Br – R0 toroidal array
Br – R+1 toroidal array
Br – R-1 toroidal array
Br – R+2 toroidal array
Br – R-2 toroidal array
Br – HFS vertical array (139 deg)
Br – HFS vertical array (199 deg)
Br – HFS toroidal array (Above midplane)
Br – HFS toroidal array (Below midplane)
Br – 67 degree poloidal array (outer surface of vacuum vessel)
Br – 157 degree poloidal array (outer surface of vacuum vessel)
Br – R+1 toroidal array (outer surface of vacuum vessel)
Br – R0 toroidal array (outer surface of vacuum vessel)

Br – R-1 toroidal array (outer surface of vacuum vessel)
Br – R0 toroidal array (external)
Br – R+1 toroidal array (external)
Br – R-1 toroidal array (external)

TOROIDAL FIELD PROBES

Bt – R0 toroidal array

POLOIDAL FLUX LOOPS

Psi – On F-coils
Psi – On vacuum vessel
Psi – On lower divertor shelf

DIAMAGNETIC LOOPS

Diamagnetic Flux – On vacuum vessel
Diamagnetic Flux – On F-coil structure
Diamagnetic Flux – Bt compensation loops

ROGOWSKI LOOPS

Plasma Current	– On F-coil structure
F-coil Current	– At F-coil patch panel
C, I-coil Current	– At I-C patch panel
B-coil Current	– Various locations
E-coil Current	– Various locations

Part I: Individual Sensors

This section lists the dimensions and coordinates of the individual sensors.

A few sensors may appear more than once, when they are members of intersecting arrays. The list includes only diagnostics that are currently mounted on the machine. All signals should be available, with a few exceptions:

- The external toroidal arrays of Br sensors at R±1 (USL-- and LSL--) were not digitized from 2001 to 2019.
- Bp and Br sensors #2A-5A and #2B-5B in the 199 degree vertical arrays are not digitized individually; only the difference with the corresponding 139 degree sensor is used.
- Two of the eight Bp and Br sensors in the HFS below-midplane toroidal arrays are digitized individually; the others are used only as members of differential pairs.
- Two of the four Bp and Br sensors in each of the R±2 toroidal arrays are digitized individually; the others are used only as members of differential pairs.
- Most of the Bp probes in the R0 toroidal array and 322 degree poloidal array are digitized as both integrated (B) and unintegrated (dB/dt) signals.
- Some of the Bp probes in the toroidal arrays at R0 and the lower divertor are digitized only as dB/dt, not integrated.

In the tables below:

R and Z	are the radial and vertical coordinates of the center of the loop (Bp, Bt, Br), or of the loop conductor itself (Psi).
Φ	is the toroidal coordinate of the center of the loop, in DIII-D's *clockwise* "machine angle" system.
γ	is the inclination angle of the sensor's axis (i.e. the direction of the field component measured by the sensor) relative to horizontal, with rotation of the axis in the R-Z plane (Bp and Br), or in the Φ🎗💐 plane (Bt).
L	is the sensor's length parallel to the measurement axis (Bp and Bt), or perpendicular to the axis and in the poloidal direction (Br).
W	is the sensor's width transverse to the axis in the toroidal direction (Bp and Br), or in the poloidal direction (Bt).
NA	is the effective area (turns-area product) of the loop (Bp, Bt, Br). The

All of these parameters are listed for the localized sensors (Bp, Bt, Br).
For the poloidal flux loops, only the R and Z coordinates are relevant.
For the Rogowski loops, only the mutual inductance with respect to the measured current is relevant. This parameter is not included here, but could be added later.

Naming Conventions

The poloidally and toroidally resolved magnetic sensors on and near the vacuum vessel have names that are intended to convey the type of measurement and the approximate location of the sensors. With some exceptions, the names follow the convention described below.

The name is divided into three fields. (The third is omitted for axisymmetric flux loops.)

Name = (Type) (Poloidal location) (Toroidal location)

Where the three fields are encoded as follows:

(Type): Two or three letters designating the type of sensor.

MPI	Magnetic probe inside the vessel, measuring tangential (poloidal) field
BTI	Magnetic probe inside the vessel, measuring toroidal field
ISL	Saddle loop inside the vessel, measuring normal (radial) field
ESL	Saddle loop external to the vessel and F-coils, measuring normal field
SL	Saddle loop on the exterior surface of the vessel, measuring normal field
DSL	(1) Saddle loop on exterior surface of the vessel, measuring normal field (2) Saddle loop on an upper divertor baffle, measuring normal field
PSI	(1) Poloidal flux (psi) loop on the exterior surface of the vessel (2) Poloidal flux (psi) loop on the divertor shelf
PSF	Poloidal flux (psi) loop on an F-coil

(Poloidal location): One or two numbers followed by one or two letters, designating the approximate poloidal location. Sensors on the vacuum vessel and sensors on the divertor baffles use slightly different systems.

Vacuum vessel wall:

Nearest F-coil	One or two digits giving the number of the nearest F-coil, or the two nearest F-coils for a sensor located between coils. Exception: Magnetic probes and saddle loops in the HFS vertical arrays use sequence numbers 1, 2, … that follow the order but not the positions of nearby F-coils.
N/F	Letter indicating Nearer to or Farther from the midplane, if needed for two sensors with otherwise similar locations and names.
A/B/M	Letter indicating Above, Below, or at the Midplane.

Divertor baffles:

Number	Magnetic probes, saddle loops and flux loops on divertor baffles use sequence numbers 1, 2, … that are unrelated to F-coils.
U/L	Letter indicating Upper or Lower divertor

(Toroidal location): Three digits (with a leading zero if needed) designating the nominal toroidal angle, in degrees of clockwise "machine angle."

Note: The three-digit angle is only approximate. Most are rounded to the nearest degree, but some may differ from the actual position by up to 7.5 degrees, for historical reasons or in order to maintain similar names within a poloidal array.

Examples (spaces are inserted into the names for clarity)

MPI 1A 322	Poloidal field probe near the F1A coil, at ϕ=322º toroidal angle
MPI 6NA 322	Poloidal field probe near the F6A coil at ϕ=322º, the nearer one to the midplane of two such probes
MPI 66M 097	Poloidal field probe at the outboard midplane, at ϕ=97º
ISL 67A 132	Saddle loop in the vessel, between F6A and F7A coils, centered at at ϕ=132º
PSF 5B	Flux loop on the F5B coil.
PSF 7FA	Flux loop on the upper (farther from the midplane) corner of the F7A coil
PSI 34B	Flux loop on the vacuum vessel, between the F3B and F4B coils

Alternate Names:

The older "3D" arrays (6 each) of magnetic probes at R±1, external saddle loops at R0, and internal saddle loops at R0 and R±1 were originally given simplified ptdata pointnames that do not follow these naming conventions described here.

After the 2013 upgrade for "3D" magnetics, these sensors were assigned MDSplus tags that follow the naming conventions. The physics data was accessible under both names.

In 2020, most of the simplified names were dropped, and the ptdata pointnames were changed to the standardized names. The remaining exceptions are the original low field side, midplane ESL* and ESLD* names.

In the tables below, the first column gives the standardized names.

- In most cases, the name in the first column is the ptdata pointname, and the second column shows the former ptdata pointname (if any) for historical information.
- In the case of low field side, midplane ESL* and ESLD* names, if the name in the first column is an MDSplus tag, then the second column shows the ptdata pointname.

NOTE: The MDSplus names provide data only in physics units. For older data, if units of volts or digitizer counts are needed, then the ptdata pointname must be used.

Unfortunately and confusingly, after the simplified names were dropped in 2020, many applications continue to reference the standardized names as MDSplus tags, with physics units only. In recent shots, data for these signals with units of volts or digitizer counts can be obtained with a direct ptdata call, but are not accessble with gadat (for example).

POLOIDAL FIELD PROBES

Bp - 322 degree poloidal array

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
MPI11M322		0.973	-0.002	322.5	89.9	0.115	0.0556	0.04977
MPI1A322		0.974	0.182	322.5	90.0	0.140	0.0556	0.05798
MPI2A322		0.974	0.512	322.5	89.7	0.140	0.0556	0.05929
MPI3A322		0.975	0.850	322.5	90.2	0.140	0.0556	0.05843
MPI4A322		0.972	1.161	322.5	90.5	0.140	0.0556	0.05851
MPI5A322		1.051	1.330	322.5	44.6	0.141	0.0556	0.05891
MPI8A322		1.219	1.406	322.5	0.3	0.139	0.0556	0.06072
MPI89A322		1.402	1.407	322.5	0.7	0.115	0.0556	0.04833
MPI9A322		1.584	1.408	322.5	0.6	0.140	0.0556	0.05917
MPI79FA322		1.783	1.323	322.5	-39.3	0.141	0.0556	0.05963
MPI79NA322		1.924	1.206	322.5	-39.2	0.114	0.0556	0.04792
MPI7FA322		2.067	1.090	322.5	-39.4	0.140	0.0556	0.05700
MPI7NA322		2.219	0.870	323.3	-68.1	0.140	0.0556	0.05863
MPI67A322		2.270	0.746	321.7	-67.9	0.141	0.0556	0.05998
MPI6FA322		2.319	0.623	323.2	-68.0	0.140	0.0556	0.05768
MPI6NA322		2.416	0.249	317.4	-89.3	0.141	0.0556	0.05958
MPI66M322		2.418	-0.001	317.4	-89.9	0.140	0.0556	0.06040
MPI1B322		0.974	-0.187	322.5	90.1	0.140	0.0556	0.06017
MPI2B322		0.975	-0.512	322.5	90.3	0.140	0.0556	0.05962
MPI3B322		0.974	-0.854	322.5	89.8	0.140	0.0556	0.05960
MPI4B322		0.972	-1.159	322.5	89.6	0.141	0.0556	0.05856
MPI5B322		1.048	-1.330	322.5	136.4	0.140	0.0556	0.06048
MPI8B322		1.254	-1.405	322.5	-180.0	0.140	0.0556	0.05967
MPI89B322		1.477	-1.406	322.5	-179.9	0.140	0.0556	0.05817
MPI9B322		1.699	-1.406	322.5	179.9	0.142	0.0556	0.05849
MPI79B322		1.894	-1.333	322.5	-129.5	0.141	0.0556	0.05751
MPI7FB322		2.085	-1.102	322.5	-129.2	0.140	0.0556	0.05773
MPI7NB322		2.212	-0.873	323.4	-113.5	0.141	0.0556	0.05741
MPI67B322		2.263	-0.749	321.7	-112.8	0.140	0.0556	0.06145
MPI6FB322		2.315	-0.624	323.3	-113.4	0.140	0.0556	0.05904
MPI6NB322		2.416	-0.244	317.4	-90.8	0.140	0.0556	0.06041

Bp - 67 degree backup probes

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
MPI2A067		0.973	0.518	75.4	89.8	0.140	0.0556	0.06015
MPI11M067		0.973	-0.004	75.4	90.0	0.141	0.0556	0.06095
MPI2B067		0.972	-0.517	75.4	90.3	0.139	0.0556	0.05997
MPI67A097	MPI67A067	2.266	0.758	97.4	-67.6	0.155	0.1000	0.13076
MPI66M067		2.413	0.003	67.5	-89.9	0.141	0.0556	0.06218
MPI67B097	MPI67B067	2.262	-0.751	97.4	-112.6	0.156	0.1000	0.13311

Bp - 142 degree poloidal array

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
MPI1A139		0.979	0.070	142.5	90.3	0.142	0.0556	0.09776
MPI2A139		0.977	0.209	135.0	89.9	0.141	0.0556	0.09808
MPI3A139		0.979	0.347	142.5	90.1	0.141	0.0556	0.09725
MPI4A139		0.976	0.485	135.0	90.0	0.141	0.0556	0.09767
MPI5A139		0.979	0.759	142.5	89.7	0.141	0.0556	0.09756
MPI79A147		1.762	1.312	147.2	-39.5	0.153	0.1000	0.21266
MPI67A142		2.262	0.757	142.7	-67.3	0.153	0.1000	0.13527
MPI67A157		2.264	0.760	157.5	-67.8	0.156	0.1000	0.13329
MPI6NA132		2.409	0.277	132.7	-89.0	0.138	0.0556	0.06170
MPI6NA157		2.407	0.254	157.6	-89.4	0.141	0.0556	0.05951
MPI66M157		2.413	-0.001	157.6	-89.8	0.137	0.0556	0.06109
MPI6NB157		2.414	-0.244	157.6	-89.6	0.141	0.0556	0.06336
MPI6FB142		2.312	-0.620	143.4	-113.0	0.140	0.0556	0.06147
MPI67B157		2.260	-0.754	157.4	-112.8	0.155	0.1000	0.13229
MPI7NB142		2.209	-0.867	143.3	-112.6	0.142	0.0556	0.06023
MPI79B142		2.044	-1.110	142.4	-129.1	0.154	0.1000	0.20995
MPI5B139		0.977	-0.757	134.9	89.8	0.142	0.0556	0.09822
MPI4B139		0.980	-0.482	142.5	89.9	0.141	0.0556	0.09748
MPI3B139		0.977	-0.343	135.0	90.1	0.141	0.0556	0.09816
MPI2B139		0.979	-0.207	142.5	89.6	0.141	0.0556	0.09812
MPI1B139		0.977	-0.069	135.0	90.0	0.142	0.0556	0.09748
MPI1B157		0.972	-0.183	157.5	89.6	0.141	0.0556	0.06068

Bp - upper divertor baffles

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
MPI1U157		1.445	1.277	155.0	-40.6	0.027	0.0556	0.01198
MPI2U157		1.560	1.187	155.0	-40.8	0.027	0.0556	0.01227
MPI3U157		1.723	1.112	155.0	-3.3	0.027	0.0556	0.01229
MPI4U157		1.873	1.116	155.4	-2.8	0.054	0.0556	0.02398
MPI5U157		1.218	1.298	152.2	66.3	0.027	0.0556	0.01139
MPI6U157		1.075	1.202	152.2	0.7	0.027	0.0556	0.01154
MPI7U157		0.976	0.979	157.5	90.3	0.027	0.0556	0.01144

Br - upper divertor baffles

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
DSL1U180		1.506	1.270	185.0	49.5	0.094	60.1	0.13912
DSL2U180		1.590	1.198	185.0	49.5	0.088	60.2	0.14843
DSL3U180		1.730	1.142	185.2	92.3	0.106	59.2	0.18909
DSL4U157		1.877	1.133	157.6	90.5	0.196	6.0	0.07450
DSL5U157		1.185	1.286	168.7	156.6	0.091	108.0	0.20466
DSL6U157		1.086	1.228	168.7	90.4	0.107	107.6	0.21782

Bp - R0 ELM array (127-137 deg.)

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
MPI66M127		2.413	0.002	127.9	-90.0	0.140	0.0556	0.05984
MPI66M132		2.409	0.007	132.5	-90.3	0.055	0.0556	0.02387
MPI66M137		2.416	0.006	137.4	-90.4	0.054	0.0556	0.02310
MPI66B137		2.410	-0.124	137.5	-89.8	0.053	0.0556	0.02448
MPI6NB137		2.409	-0.255	137.5	-90.5	0.053	0.0556	0.02378

Bp - R0 ELM array (307-317 deg)

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
MPI66M307		2.413	0.001	307.0	-90.2	0.140	0.0556	0.05984
MPI66M312		2.411	-0.001	312.4	-90.0	0.054	0.0556	0.02297
MPI6NA312		2.411	0.264	313.1	-89.2	0.055	0.0556	0.02419
MPI66B312		2.411	-0.131	313.2	-90.3	0.055	0.0556	0.02398
MPI6NB312		2.411	-0.261	313.2	-90.3	0.055	0.0556	0.02394
MPI66M322		2.418	-0.001	317.4	-89.9	0.140	0.0556	0.06040

Bp - lower divertor toroidal array

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
MPI1L020		1.5070	-1.2885	20.	-178.17	0.0271	0.0556	0.01207
MPI2L020		1.6064	-1.2892	20.	-179.69	0.0274	0.0556	0.01212
MPI1L050		1.5055	-1.2883	50.	-179.73	0.0269	0.0556	0.01198
MPI1L110		1.5127	-1.2900	110.	-178.96	0.0279	0.0556	0.01226
MPI1L180		1.5104	-1.2915	180.	-179.99	0.0271	0.0556	0.01130
MPI2L180		1.6103	-1.2915	180.	-179.75	0.0277	0.0556	0.01117
MPI3L180		1.7103	-1.2918	180.	-178.98	0.0273	0.0556	0.01125
MPI1L230		1.5070	-1.2920	230.	179.61	0.0271	0.0556	0.01103
MPI1L320		1.5055	-1.2899	320.	179.74	0.0273	0.0556	0.01194

Bp - R0 toroidal array

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
MPI66M020		2.410	0.002	19.5	-89.9	0.139	0.0556	0.06202
MPI66M067		2.413	0.003	67.5	-89.9	0.141	0.0556	0.06218
MPI66M097		2.413	-0.005	97.4	-89.8	0.137	0.0556	0.06087
MPI66M127		2.413	0.002	127.9	-90.0	0.140	0.0556	0.05984
MPI66M132		2.409	0.007	132.5	-90.3	0.055	0.0556	0.02387
MPI66M137		2.416	0.006	137.4	-90.4	0.054	0.0556	0.02310
MPI66M157		2.413	-0.001	157.6	-89.8	0.137	0.0556	0.06109
MPI66M200		2.412	0.003	199.7	-89.9	0.139	0.0556	0.06048
MPI66M247		2.413	-0.003	246.4	-90.5	0.140	0.0556	0.06150
MPI66M277		2.413	-0.009	277.5	-89.7	0.137	0.0556	0.06194
MPI66M307		2.413	0.001	307.0	-90.2	0.140	0.0556	0.05984
MPI66M312		2.411	-0.001	312.4	-90.0	0.054	0.0556	0.02297
MPI66M322		2.418	-0.001	317.4	-89.9	0.140	0.0556	0.06040
MPI66M340		2.413	-0.002	339.7	-89.8	0.140	0.0556	0.06072

Bp - R+1 toroidal array

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
MPI67A022		2.273	0.767	22.3	-67.5	0.153	0.1000	0.14410
MPI67A037	MPI67A1	2.268	0.755	37.5	-67.9	0.155	0.1000	0.13018
MPI67A052		2.270	0.764	52.4	-65.9	0.154	0.1000	0.14439
MPI67A067		2.261	0.758	67.5	-68.0	0.153	0.1000	0.13683
MPI67A082		2.260	0.757	82.5	-67.6	0.154	0.1000	0.13749
MPI67A097	MPI67A2	2.266	0.758	97.4	-67.6	0.155	0.1000	0.13076
MPI67A142		2.262	0.757	142.7	-67.3	0.153	0.1000	0.13527
MPI67A157	MPI67A3	2.264	0.760	157.5	-67.8	0.156	0.1000	0.13329
MPI67A217	MPI67A4	2.268	0.753	217.5	-67.6	0.155	0.1000	0.13148
MPI67A262		2.264	0.751	262.6	-67.5	0.154	0.1000	0.13937
MPI67A277	MPI67A5	2.267	0.756	277.3	-68.0	0.155	0.1000	0.13266
MPI67A307		2.266	0.760	307.5	-67.5	0.152	0.1000	0.13519
MPI67A337	MPI67A6	2.268	0.756	337.4	-67.6	0.155	0.1000	0.13068

Bp - R-1 toroidal array

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
MPI67B022		2.253	-0.751	22.4	-112.3	0.154	0.1000	0.14287
MPI67B037	MPI67B1	2.264	-0.752	37.4	-112.4	0.156	0.1000	0.13078
MPI67B052		2.257	-0.751	52.4	-112.7	0.154	0.1000	0.14125
MPI67B097	MPI67B2	2.262	-0.751	97.4	-112.6	0.156	0.1000	0.13311
MPI67B157	MPI67B3	2.260	-0.754	157.4	-112.8	0.155	0.1000	0.13229
MPI67B217	MPI67B4	2.266	-0.753	217.6	-112.8	0.155	0.1000	0.13554
MPI67B277	MPI67B5	2.260	-0.756	277.5	-112.5	0.155	0.1000	0.13036
MPI67B337	MPI67B6	2.260	-0.754	337.4	-112.7	0.155	0.1000	0.14188

Bp - R+2 toroidal array

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
MPI79A072		1.762	1.312	72.2	-39.0	0.153	0.1000	0.21191
MPI79A147		1.762	1.312	147.2	-39.5	0.153	0.1000	0.21266
MPI79A222		1.762	1.312	222.1	-39.5	0.153	0.1000	0.21037
MPI79A272		1.763	1.311	272.9	-39.1	0.153	0.1000	0.21067

Bp – R-2 toroidal array

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
MPI79B067		2.039	-1.110	67.5	-129.4	0.153	0.1000	0.21217
MPI79B142		2.044	-1.110	142.4	-129.1	0.154	0.1000	0.20995
MPI79B217		2.046	-1.109	217.5	-129.6	0.153	0.1000	0.21082
MPI79B277		2.043	-1.110	277.6	-129.2	0.153	0.1000	0.21190

Bp – HFS vertical array (139 deg)

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
MPI5A139		0.979	0.759	142.5	89.7	0.141	0.0556	0.09756
MPI4A139		0.976	0.485	135.0	90.0	0.141	0.0556	0.09767
MPI3A139		0.979	0.347	142.5	90.1	0.141	0.0556	0.09725
MPI2A139		0.977	0.209	135.0	89.9	0.141	0.0556	0.09808
MPI1A139		0.979	0.070	142.5	90.3	0.142	0.0556	0.09776
MPI1B139		0.977	-0.069	135.0	90.0	0.142	0.0556	0.09748
MPI2B139		0.979	-0.207	142.5	89.6	0.141	0.0556	0.09812
MPI3B139		0.977	-0.343	135.0	90.1	0.141	0.0556	0.09816
MPI4B139		0.980	-0.482	142.5	89.9	0.141	0.0556	0.09748
MPI5B139		0.977	-0.757	134.9	89.8	0.142	0.0556	0.09822

Bp – HFS vertical array (199 deg)

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
MPI5A199		0.978	0.759	202.6	90.0	0.141	0.0556	0.09805
MPI4A199		0.976	0.485	195.0	90.2	0.141	0.0556	0.09741
MPI3A199		0.977	0.347	202.6	90.0	0.141	0.0556	0.09803
MPI2A199		0.976	0.209	195.0	89.8	0.141	0.0556	0.09762
MPI1A199		0.976	0.070	202.6	90.6	0.141	0.0556	0.09796
MPI1B199		0.976	-0.069	195.0	90.3	0.141	0.0556	0.09842
MPI2B199		0.977	-0.207	202.6	89.8	0.141	0.0556	0.09740
MPI3B199		0.976	-0.344	195.0	90.0	0.142	0.0556	0.09782
MPI4B199		0.978	-0.483	202.5	89.7	0.142	0.0556	0.09836
MPI5B199		0.977	-0.757	194.9	90.0	0.142	0.0556	0.09831

Bp - HFS toroidal array (Above midplane)

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
MPI1A011		0.977	0.071	15.1	90.3	0.141	0.0556	0.09765
MPI1A049		0.979	0.070	52.6	90.2	0.141	0.0556	0.09753
MPI1A109		0.977	0.070	112.5	90.4	0.141	0.0556	0.09834
MPI1A139		0.979	0.070	142.5	90.3	0.142	0.0556	0.09776
MPI1A199		0.976	0.070	202.6	90.6	0.141	0.0556	0.09796
MPI1A244		0.977	0.070	247.3	89.9	0.141	0.0556	0.09751
MPI1A274		0.978	0.071	277.5	90.3	0.141	0.0556	0.09771
MPI1A341		0.978	0.071	345.1	89.9	0.141	0.0556	0.09756

Bp - HFS toroidal array (Below midplane)

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
MPI1B011		0.976	-0.068	7.6	90.1	0.141	0.0556	0.09785
MPI1B049		0.978	-0.068	45.1	90.0	0.141	0.0556	0.09860
MPI1B109		0.976	-0.069	105.0	89.8	0.141	0.0556	0.09854
MPI1B139		0.977	-0.069	135.0	90.0	0.142	0.0556	0.09748
MPI1B199		0.976	-0.069	195.0	90.3	0.141	0.0556	0.09842
MPI1B244		0.977	-0.069	239.9	90.3	0.141	0.0556	0.09777
MPI1B274		0.976	-0.070	270.1	90.2	0.141	0.0556	0.09762
MPI1B341		0.977	-0.069	337.6	89.8	0.141	0.0556	0.09755

RADIAL FIELD LOOPS

Br - R0 toroidal array

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
ISL66M017	MISL1	2.431	0.000	17.2	0.0	0.800	60.3	2.0466
ISL66M042		2.431	0.000	40.5	0.0	0.800	56.0	1.9040
ISL66M072	MISL2	2.431	0.000	72.4	0.0	0.800	50.1	1.7007
ISL66M102		2.431	0.000	100.2	0.0	0.800	63.3	2.1522
ISL66M132	MISL3	2.431	0.000	132.6	0.0	0.800	70.3	2.3850
ISL66M197	MISL4	2.431	0.000	197.6	0.0	0.800	59.6	2.0238
ISL66M252	MISL5	2.431	0.000	252.4	0.0	0.800	50.1	1.7007
ISL66M312	MISL6	2.431	0.000	312.3	0.0	0.800	69.6	2.3633

Br - R+1 toroidal array

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
ISL67A017	UISL1	2.300	0.714	17.8	22.6	0.680	61.5	1.6779
ISL67A052		2.300	0.714	50.1	22.6	0.680	60.0	1.6371
ISL67A072	UISL2	2.300	0.714	73.0	22.6	0.680	48.9	1.3344
ISL67A112		2.300	0.714	111.3	22.6	0.680	62.4	1.7021
ISL67A132	UISL3	2.300	0.714	133.0	22.6	0.680	71.1	1.9396
ISL67A197	UISL4	2.300	0.714	198.6	22.6	0.680	60.0	1.6370
ISL67A252	UISL5	2.300	0.714	253.0	22.6	0.680	48.9	1.3344
ISL67A312	UISL6	2.300	0.714	312.3	22.6	0.680	69.6	1.8997

Br - R-1 toroidal array

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
ISL67B017	LISL1	2.300	-0.714	18.7	-22.6	0.680	59.7	1.6289
ISL67B052		2.300	-0.714	50.1	-22.6	0.680	60.0	1.6371
ISL67B072	LISL2	2.300	-0.714	73.0	-22.6	0.680	48.9	1.3344
ISL67B112		2.300	-0.714	111.4	-22.6	0.680	62.5	1.7058
ISL67B132	LISL3	2.300	-0.714	133.0	-22.6	0.680	71.1	1.9396
ISL67B197	LISL4	2.300	-0.714	198.6	-22.6	0.680	60.0	1.6370
ISL67B252	LISL5	2.300	-0.714	253.0	-22.6	0.680	48.9	1.3344
ISL67B312	LISL6	2.300	-0.714	313.2	-22.6	0.680	71.4	1.9486

Br - R+2 toroidal array

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
ISL79A072		1.773	1.326	72.2	51.0	0.206	2.82	0.56651
ISL79A147		1.773	1.327	147.2	51.0	0.206	2.82	0.56990
ISL79A222		1.773	1.327	222.1	51.0	0.206	2.82	0.55330
ISL79A272		1.774	1.325	272.9	51.0	0.206	2.82	0.57139

Br – R-2 toroidal array

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
ISL79B067		2.053	-1.122	67.5	-40.1	0.206	2.43	0.56541
ISL79B142		2.058	-1.122	142.4	-40.1	0.206	2.43	0.56309
ISL79B217		2.060	-1.121	217.5	-40.1	0.206	2.43	0.56656
ISL79B277		2.057	-1.122	277.6	-40.1	0.206	2.43	0.57099

Br – HFS vertical array (139 deg)

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
ISL5A139		0.959	0.759	138.7	179.9	0.116	20.9	0.24262
ISL4A139		0.959	0.486	138.8	180.0	0.116	20.9	0.24262
ISL3A139		0.960	0.348	138.8	180.0	0.116	20.9	0.24262
ISL2A139		0.960	0.209	138.7	179.7	0.116	20.9	0.24262
ISL1A139		0.961	0.070	138.7	179.9	0.116	20.9	0.24262
ISL1B139		0.960	-0.071	138.7	180.0	0.116	20.9	0.24262
ISL2B139		0.960	-0.208	138.7	179.9	0.116	20.9	0.24262
ISL3B139		0.959	-0.345	138.7	179.9	0.116	20.9	0.24262
ISL4B139		0.960	-0.483	138.7	179.9	0.116	20.9	0.24262
ISL5B139		0.960	-0.762	131.1	180.0	0.116	20.9	0.24262

Br – HFS vertical array (199 deg)

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
ISL5A199		0.959	0.758	198.8	179.8	0.116	20.9	0.24262
ISL4A199		0.959	0.486	198.8	180.2	0.116	20.9	0.24262
ISL3A199		0.959	0.347	198.8	179.9	0.116	20.9	0.24262
ISL2A199		0.959	0.209	198.7	179.9	0.116	20.9	0.24262
ISL1A199		0.960	0.070	198.8	180.0	0.116	20.9	0.24262
ISL1B199		0.959	-0.071	198.8	179.9	0.116	20.9	0.24262
ISL2B199		0.959	-0.208	198.8	180.0	0.116	20.9	0.24262
ISL3B199		0.959	-0.345	198.8	179.9	0.116	20.9	0.24262
ISL4B199		0.959	-0.483	198.7	180.0	0.116	20.9	0.24262
ISL5B199		0.959	-0.762	198.7	180.1	0.116	20.9	0.24262

Br - HFS toroidal array (Above midplane)

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
ISL1A011		0.961	0.070	11.3	179.9	0.116	20.9	0.24262
ISL1A049		0.960	0.070	48.6	180.3	0.116	20.9	0.24262
ISL1A109		0.959	0.070	108.7	180.2	0.116	20.9	0.24262
ISL1A139		0.961	0.070	138.7	179.9	0.116	20.9	0.24262
ISL1A199		0.960	0.070	198.8	180.0	0.116	20.9	0.24262
ISL1A244		0.961	0.070	243.6	179.4	0.116	20.9	0.24262
ISL1A274		0.960	0.070	273.8	180.0	0.116	20.9	0.24262
ISL1A341		0.960	0.070	341.3	179.5	0.116	20.9	0.24262

Br - HFS toroidal array (Below midplane)

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
ISL1B011		0.961	-0.069	11.4	180.5	0.116	20.9	0.24262
ISL1B049		0.960	-0.070	48.6	179.8	0.116	20.9	0.24262
ISL1B109		0.958	-0.070	108.7	179.8	0.116	20.9	0.24262
ISL1B139		0.960	-0.071	138.7	180.0	0.116	20.9	0.24262
ISL1B199		0.959	-0.071	198.8	179.9	0.116	20.9	0.24262
ISL1B244		0.961	-0.069	243.6	180.6	0.116	20.9	0.24262
ISL1B274		0.960	-0.070	273.7	180.2	0.116	20.9	0.24262
ISL1B341		0.960	-0.069	341.3	180.3	0.116	20.9	0.24262

Br - 67 degree poloidal array (outer surface of vacuum vessel)

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
DSL12A067		0.925	0.343	67.0	180.0	0.682	30.0	0.33011
DSL34A067		0.942	1.003	67.0	176.9	0.631	30.0	0.31130
DSL59A067		1.320	1.393	67.0	101.4	0.729	29.7	0.49906
DSL79A067		1.946	1.263	67.0	51.0	0.639	29.6	0.64262
DSL67A067		2.315	0.780	67.0	22.4	0.557	29.7	0.66875
DSL66M052		2.433	0.000	52.0	-0.1	1.036	24.8	1.09124
DSL67B067		2.313	-0.781	67.0	-22.4	0.530	29.9	0.64055
DSL79B067		2.025	-1.262	67.0	-39.9	0.519	29.9	0.54883
DSL59B067		1.405	-1.392	67.0	-99.1	0.896	29.9	0.65748
DSL34B067		0.942	-1.003	67.0	-176.9	0.631	30.0	0.31130
DSL12B067		0.925	-0.343	67.0	180.0	0.682	30.0	0.33011

Br - 157 degree poloidal array (outer surface of vacuum vessel)

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
DSL12A157		0.925	0.343	157.0	180.0	0.682	30.0	0.33011
DSL34A157		0.942	1.003	157.0	176.9	0.631	30.0	0.31130
DSL59A157		1.320	1.393	157.0	101.4	0.729	30.0	0.50359
DSL79A157		1.944	1.265	157.0	51.0	0.635	29.9	0.64307
DSL67A157		2.314	0.783	157.0	22.4	0.557	30.1	0.67718
DSL66M152		2.433	0.001	152.0	0.0	1.037	29.1	1.28213
DSL67B157		2.313	-0.781	157.0	-22.4	0.549	30.0	0.66533
DSL79B157		2.025	-1.262	157.0	-39.9	0.523	30.0	0.55500
DSL59B157		1.406	-1.392	157.0	-99.1	0.897	30.0	0.65995
DSL34B157		0.942	-1.003	157.0	-176.9	0.631	30.0	0.31130
DSL12B157		0.925	-0.343	157.0	180.0	0.682	30.0	0.33011

Br - R+1 toroidal array (outer surface of vacuum vessel)

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
DSL67A067		2.315	0.780	67.0	22.4	0.557	29.7	0.66875
DSL67A157		2.314	0.783	157.0	22.4	0.557	30.1	0.67718
SL67FA345		2.289	0.845	345.0	22.4	0.432	21.0	0.36133
SL67NA345		2.395	0.587	345.0	22.4	0.124	21.0	0.10858

Br - R0 toroidal array (outer surface of vacuum vessel)

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
DSL66M052		2.433	0.000	52.0	-0.1	1.036	24.8	1.09124
SL66A132		2.449	0.261	132.0	3.5	0.519	13.1	0.28958
SL66B132		2.447	-0.262	132.0	-3.8	0.522	13.1	0.29140
DSL66M152		2.433	0.001	152.0	0.0	1.037	29.1	1.28213
SL66A312		2.449	0.260	312.0	3.4	0.519	12.8	0.28481
SL66B312		2.447	-0.261	312.0	-3.7	0.522	12.8	0.28627

Br - R-1 toroidal array (outer surface of vacuum vessel)

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
SL67NB015		2.393	-0.588	15.0	-22.4	0.124	15.0	0.07765
SL67FB015		2.287	-0.845	15.0	-22.4	0.430	15.0	0.25751
DSL67B067		2.313	-0.781	67.0	-22.4	0.530	29.9	0.64055
DSL67B157		2.313	-0.781	157.0	-22.4	0.549	30.0	0.66533

Br - R0 toroidal array (external)

NAME	Present PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
ESL66M019	ESL019	2.477	0.000	19.0	0.0	1.194	60.0	3.0960
ESL66M079	ESL079	2.477	0.000	79.0	0.0	1.194	60.0	3.0960
ESL66M139	ESL139	2.477	0.000	139.0	0.0	1.194	60.0	3.0960
ESL66M199	ESL199	2.477	0.000	199.0	0.0	1.194	60.0	3.0960
ESL66M259	ESL259	2.477	0.000	259.0	0.0	1.194	60.0	3.0960
ESL66M319	ESL319	2.477	0.000	319.0	0.0	1.194	60.0	3.0960

Br - R+1 toroidal array (external)

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
ESL67A004	USL1	2.389	0.806	4.0	24.1	0.454	30.0	2.2697
ESL67A034	USL2	2.389	0.806	34.0	24.1	0.454	30.0	2.2697
ESL67A064	USL3	2.389	0.806	64.0	24.1	0.454	30.0	2.2697
ESL67A094	USL4	2.389	0.806	94.0	24.1	0.454	30.0	2.2697
ESL67A124	USL5	2.389	0.806	124.0	24.1	0.454	30.0	2.2697
ESL67A154	USL6	2.389	0.806	154.0	24.1	0.454	30.0	2.2697
ESL67A184	USL7	2.389	0.806	184.0	24.1	0.454	30.0	2.2697
ESL67A214	USL8	2.389	0.806	214.0	24.1	0.454	30.0	2.2697
ESL67A244	USL9	2.389	0.806	244.0	24.1	0.454	30.0	2.2697
ESL67A274	USL10	2.389	0.806	274.0	24.1	0.454	30.0	2.2697
ESL67A304	USL11	2.389	0.806	304.0	24.1	0.454	30.0	2.2697
ESL67A334	USL12	2.389	0.806	334.0	24.1	0.454	30.0	2.2697

Br - R-1 toroidal array (external)

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
ESL67B004	LSL1	2.389	-0.806	4.0	-24.1	0.454	30.0	2.2697
ESL67B034	LSL2	2.389	-0.806	34.0	-24.1	0.454	30.0	2.2697
ESL67B064	LSL3	2.389	-0.806	64.0	-24.1	0.454	30.0	2.2697
ESL67B094	LSL4	2.389	-0.806	94.0	-24.1	0.454	30.0	2.2697
ESL67B124	LSL5	2.389	-0.806	124.0	-24.1	0.454	30.0	2.2697
ESL67B154	LSL6	2.389	-0.806	154.0	-24.1	0.454	30.0	2.2697
ESL67B184	LSL7	2.389	-0.806	184.0	-24.1	0.454	30.0	2.2697
ESL67B214	LSL8	2.389	-0.806	214.0	-24.1	0.454	30.0	2.2697
ESL67B244	LSL9	2.389	-0.806	244.0	-24.1	0.454	30.0	2.2697
ESL67B274	LSL10	2.389	-0.806	274.0	-24.1	0.454	30.0	2.2697
ESL67B304	LSL11	2.389	-0.806	304.0	-24.1	0.454	30.0	2.2697
ESL67B334	LSL12	2.389	-0.806	334.0	-24.1	0.454	30.0	2.2697

TOROIDAL FIELD PROBES

Bt - R0 toroidal array

NAME	former PTDATA name	POSITION				SIZE		
		R (m)	Z (m)	Φ (deg)	γ (deg)	L (m)	W (m)	NA(m^2)
BTI66M053		2.414	-0.271	52.6	0.1	0.140	0.0556	0.06130
BTI66M132		2.414	0.145	129.8	0.3	0.139	0.0556	0.05943
BTI66M233		2.414	-0.276	232.7	0.0	0.140	0.0556	0.06180
BTI66M312		2.417	0.134	314.4	0.5	0.141	0.0556	0.05947

Bt - other (also see diamagnetic compensation)

POINTNAME	Phi (deg)	Location	Type	Other names	Comment
BT	97	R0	Pancake coil	CL04	Similar to BTCOMP

POLOIDAL FLUX LOOPS

Psi – On F-coils

POINTNAME	R (m)	Z (m)
PSF1A	0.8929	0.1683
PSF2A	0.8936	0.5092
PSF3A	0.8950	0.8500
PSF4A	0.8932	1.1909
PSF5A	1.0106	1.4475
PSF8A	1.2517	1.5517
PSF9A	1.6885	1.5169
PSF7FA	2.2090	1.2130
PSF7NA	2.2850	1.0180
PSF6FA	2.5010	0.5270
PSF6NA	2.5110	0.3320
PSF6NB	2.5090	-0.3350
PSF6FB	2.5010	-0.5240
PSF7NB	2.2970	-1.0100
PSF7FB	2.2150	-1.2140
PSF9B	1.6882	-1.5075
PSF8B	1.2491	-1.5527
PSF5B	1.0152	-1.4536
PSF4B	0.8952	-1.1953
PSF3B	0.8933	-0.8543
PSF2B	0.8928	-0.5135
PSF1B	0.8929	-0.1726

← PSF1A is the reference flux, and its signal represents the total poloidal flux at its location. All other PSF and PSI flux loops are connected differentially with respect to PSF1A, and their signal represents the flux difference.

Four measurements of total (non-differential) poloidal flux are available:

PSF1A	reference flux
PSF6NATOTL	same loop as PSF6NA
PSI11MTOTL	same loop as PSI11M
PSI6ATOTL	same loop as PSI6A

Four measurements of loop voltage are available from these four non-differential flux measurements:

VLOOP	= d/dt (PSF1A)
VLOOPF6NA	= d/dt (PSF6NATOTL)
VLOOPI11M	= d/dt (PSI11MTOTL)
VLOOPI6A	= d/dt (PSI6ATOTL)

Psi – On vacuum vessel

POINTNAME	R (m)	Z (m)
PSI11M	0.9247	0.0000
PSI12A	0.9247	0.3429
PSI23A	0.9247	0.6858
PSI34A	0.9247	1.0287
PSI45A	0.9621	1.3208
PSI58A	1.1212	1.4646
PSI9A	1.6978	1.4646
PSI7A	2.1933	1.0629
PSI6A	2.4347	0.4902
PSI6B	2.4328	-0.4909
PSI7B	2.1933	-1.0602
PSI9B	1.8559	-1.4625
PSI89NB	1.5771	-1.4625
PSI89FB	1.3828	-1.4625
PSI58B	1.1199	-1.4625
PSI45B	0.9611	-1.3198
PSI34B	0.9247	-1.0287
PSI23B	0.9247	-0.6858
PSI12B	0.9247	-0.3429

Psi – On lower divertor shelf

POINTNAME	R (m)	Z (m)
PSI1L	1.3929	-1.3057
PSI2L	1.5983	-1.3057
PSI3L	1.7759	-1.3053

DIAMAGNETIC LOOPS

Diamag – On vacuum vessel

POINTNAME	Phi (deg)	Comment				
DIAMAG1	280.5	Original name DIO4A				
DIO3A	100.5	Spare loop				
DIO3B	100.5	Spare loop				
DIO4B	280.5	Spare loop				

Diamag – On F-coil structure

POINTNAME	Phi (deg)	Comment				
DIAMAG2	97.5	Original name DIO5A				
DIAMAG3	277.5	Original name DIO6A				
DIO5B	97.5	Spare loop				
DIO6B	277.5	Spare loop				

Compensation loops – Various locations ex-vessel

POINTNAME	Phi (deg)	Location	Type	Other names	Comment
BTCOMP	97	R0	Pancake coil	CC1, CL03	Comp. for DIAMAG1
DIA2ROG	210	R-1	Rog around Bcoil bundle		Comp. for DIAMAG2
CC2	(0-360)	On F8B	5-turn Rog around c'post	CL02N	(Dead)
CC3	(0-360)	On F8B	4-turn Rog around c'post	CL02F	Comp. for DIAMAG3

PLASMA CURRENT ROGOWSKI LOOPS

Ip – On F-coil structure

POINTNAME	Phi (deg)	Comment				
RL01	307.5					
RL02	67.5					
RL03	127.5	Used for IP pointname				

COIL CURRENT ROGOWSKI LOOPS

F-coils – At F-coil patch panel

Above midplane	F1A	F2A	F3A	F4A	F5A	F6A	F7A	F8A	F9A
Below midplane	F1B	F2B	F3B	F4B	F5B	F6B	F7B	F8B	F9B

C and I-coils – At I-C patch panel

At midplane	C19	C79	C139	C199	C259	C319
Above midplane	IU30	IU90	IU150	IU210	IU270	IU330
Below midplane	IL30	IL90	IL150	IL210	IL270	IL330

(alternate names, fast sampling)

At midplane	C19F	C79F	C139F	C199F	C259F	C319F
Above midplane	IU30F	IU90F	IU150F	IU210F	IU270F	IU330F
Below midplane	IL30F	IL90F	IL150F	IL210F	IL270F	IL330

B-coil

POINTNAME	Phi (deg)	Location	Measures
B01		Coax in lower pit	B-supply module 1 current
B02		Coax in lower pit	B-supply module 2 current
BCOIL	30	B-coil bus	B-coil current
BCOIL2	195	B-coil bus	B-coil current

E-coil

POINTNAME	Phi (deg)	Location	Measures
ECOILA			Solenoid winding A
ECOILB			Solenoid winding B
E567UP			E49-52A, E60-61A
E567DN			E49-52B, E60-61B
E89UP			E53-59A
E89DN			E53-59B

Part II: Differential Pairs for "3D" Magnetic Arrays

This section documents the differential pair connections of the DIII-D "3D" magnetic arrays. The following arrays of sensor pairs are included:

POLOIDAL FIELD PROBES

Bp – R0 toroidal array
Bp – R+1 toroidal array
Bp – R-1 toroidal array
Bp – R+2 toroidal array
Bp – R-2 toroidal array
Bp – HFS toroidal array (Above midplane)
Bp – HFS toroidal array (Below midplane)
Bp – HFS vertical array (139-199 deg)

RADIAL FIELD LOOPS

Br – R0 toroidal array
Br – R+1 toroidal array
Br – R-1 toroidal array
Br – R+2 toroidal array
Br – R-2 toroidal array
Br – HFS toroidal array (Above midplane)
Br – HFS toroidal array (Below midplane)
Br – HFS vertical array (139-199 deg)
Br – R0 toroidal array (external)
Br – R0 toroidal array (external) – with hardware compensation

TOROIDAL FIELD PROBES

Bt – R0 toroidal array

These are the core set of arrays for "3D" magnetic field measurements. The seven toroidal arrays and two vertical arrays of poloidal field probes and radial field loops with similar locations comprise a set of toroidally and poloidally resolved 2-axis measurements. These arrays are connected in differential pairs to eliminat the axisymmetric field. A few sensor pairss are listed twice, as members of intersecting toroidal and vertical arrays.

Most of the individual sensor signals are also available, with a few exceptions:

- Bp and Br sensors #2A-5A and #2B-5B in the 199 degree vertical arrays are not digitized individually; only the difference with the corresponding 139 degree sensor is used.
- Two of the eight Bp and Br sensors in the HFS toroidal array below the midplane (MPI1B***) are digitized individually; the others are used only as members of differential pairs.
- Two of the four Bp and Br sensors in each of the R±2 toroidal arrays are digitized individually; the others are used only as members of differential pairs.

The tables below give the names and Inputs of the sensor pairs. Please also see the "Naming Conventions" section in Part I of this memo for additional details.

Name The names use the following convention:
The first three characters indicate the type of measurement.
(MPI, ISL, ESL, BTI)
"D" following these denotes a differential pair.
The next three characters indicate the general poloidal location.
The last three characters give the approximate toroidal location in degrees (in the case of a differential pair, the toroidal location of the first member).

Alternate Name is an newer MDSplus tag that follows the standard naming convention. These are provided in cases of older pairs with simpler ptdata pointnames. The data can be accessed under both names.

The tables list the standardized names in the first column, for consistency. In the cases where this name is an MDSplus tag, the older ptdata pointnae is listed in the second column.

NOTE: The MDSplus names provide data only in physics units, <u>not</u> volts or digitizer counts.

Except for the vertical arrays, each sensor is a member of two pairs. The calibration factors for the pairs are defined such that in physical units the differential signal is half the difference of (Input #1) – (Input #2). For example:

$$\Delta B(\text{MPID66M020}) = [B_1(\text{MPI66M020}) - B_2(\text{MPI66M097})] / 2.$$

Three of the external saddle loop pairs have a second version of the signal that includes hardware compensation for direct pickup from nearby coils. These are listed as a separate array.

All signals listed in this memo have units of Tesla, averaged over the active area of the sensor.

Differential pair connections

Bp – R0 toroidal array

PAIR NAME	*former PTDATA name*	*Input #1*	*Input #2*
MPID66M020		MPI66M020	MPI66M097
MPID66M067	MPID067U	MPI66M067	MPI66M247
MPID66M097	MPID097U	MPI66M097	MPI66M277
MPID66M127	MPID127U	MPI66M127	MPI66M307
MPID66M157	MPID157U	MPI66M157	MPI66M340
MPID66M200		MPI66M200	MPI66M020
MPID66M247		MPI66M247	MPI66M127
MPID66M277		MPI66M277	MPI66M157
MPID66M307		MPI66M307	MPI66M200
MPID66M340		MPI66M340	MPI66M067

Bp – R+1 toroidal array

PAIR NAME	*former PTDATA name*	*Input #1*	*Input #2*
MPID67A022		MPI67A022	MPI67A217
MPID67A037	UMPID037	MPI67A037	MPI67A067
MPID67A052		MPI67A052	MPI67A022
MPID67A067		MPI67A067	MPI67A262
MPID67A082		MPI67A082	MPI67A052
MPID67A097	UMPID097	MPI67A097	MPI67A082
MPID67A142		MPI67A142	MPI67A037
MPID67A217		MPI67A217	MPI67A097
MPID67A262		MPI67A262	MPI67A277
MPID67A277		MPI67A277	MPI67A307
MPID67A307		MPI67A307	MPI67A337
MPID67A337		MPI67A337	MPI67A142

Bp – R-1 toroidal array

PAIR NAME	*former PTDATA name*	*Input #1*	*Input #2*
MPID67B022		MPI67B022	MPI67B052
MPID67B037	LMPID037	MPI67B037	MPI67B217
MPID67B052		MPI67B052	MPI67B037
MPID67B097	LMPID097	MPI67B097	MPI67B277
MPID67B157	LMPID157	MPI67B157	MPI67B337
MPID67B217		MPI67B217	MPI67B097
MPID67B277		MPI67B277	MPI67B157
MPID67B337		MPI67B337	MPI67B022

Bp – R+2 toroidal array

PAIR NAME	*former PTDATA name*	*Input #1*	*Input #2*
MPID79A072		MPI79A072	MPI79A222
MPID79A147		MPI79A147	MPI79A072
MPID79A222		MPI79A222	MPI79A272
MPID79A272		MPI79A272	MPI79A147

Bp – R-2 toroidal array

PAIR NAME	*former PTDATA name*	*Input #1*	*Input #2*
MPID79B067		MPI79B067	MPI79B217
MPID79B142		MPI79B142	MPI79B067
MPID79B217		MPI79B217	MPI79B277
MPID79B277		MPI79B277	MPI79B142

Bp – HFS toroidal array (Above midplane)

PAIR NAME	*former PTDATA name*	*Input #1*	*Input #2*
MPID1A011		MPI1A011	MPI1A199
MPID1A049		MPI1A049	MPI1A244
MPID1A109		MPI1A109	MPI1A011
MPID1A139		MPI1A139	MPI1A341
MPID1A199		MPI1A199	MPI1A139
MPID1A244		MPI1A244	MPI1A274
MPID1A274		MPI1A274	MPI1A109
MPID1A341		MPI1A341	MPI1A049

Bp – HFS toroidal array (Below midplane)

PAIR NAME	*former PTDATA name*	*Input #1*	*Input #2*
MPID1B011		MPI1B011	MPI1B199
MPID1B049		MPI1B049	MPI1B244
MPID1B109		MPI1B109	MPI1B011
MPID1B139		MPI1B139	MPI1B341
MPID1B199		MPI1B199	MPI1B139
MPID1B244		MPI1B244	MPI1B274
MPID1B274		MPI1B274	MPI1B109
MPID1B341		MPI1B341	MPI1B049

Bp – HFS vertical array

PAIR NAME	*former PTDATA name*	*Input #1*	*Input #2*
MPID5A199		MPI5A199	MPI5A139
MPID4A199		MPI4A199	MPI4A139
MPID3A199		MPI3A199	MPI3A139
MPID2A199		MPI2A199	MPI2A139
MPID1A199		MPI1A199	MPI1A139
MPID1B199		MPI1B199	MPI1B139
MPID2B199		MPI2B199	MPI2B139
MPID3B199		MPI3B199	MPI3B139
MPID4B199		MPI4B199	MPI4B139
MPID5B199		MPI5B199	MPI5B139

Br – R0 toroidal array

PAIR NAME	*former PTDATA name*	*Input #1*	*Input #2*
ISLD66M017		ISL66M017	ISL66M042
ISLD66M042		ISL66M042	ISL66M072
ISLD66M072	ISLD079U	ISL66M072	ISL66M252
ISLD66M102		ISL66M102	ISL66M132
ISLD66M132	ISLD139U	ISL66M132	ISL66M312
ISLD66M197	ISLD199U	ISL66M197	ISL66M017
ISLD66M252		ISL66M252	ISL66M102
ISLD66M312		ISL66M312	ISL66M197

Br – R+1 toroidal array

PAIR NAME	*former PTDATA name*	*Input #1*	*Input #2*
ISLD67A017		ISL67A017	ISL67A052
ISLD67A052		ISL67A052	ISL67A072
ISLD67A072	UISLD079	ISL67A072	ISL67A252
ISLD67A112		ISL67A112	ISL67A132
ISLD67A132	UISLD139	ISL67A132	ISL67A312
ISLD67A197	UISLD199	ISL67A197	ISL67A017
ISLD67A252		ISL67A252	ISL67A112
ISLD67A312		ISL67A312	ISL67A197

Br – R-1 toroidal array

PAIR NAME	***former PTDATA name***	***Input #1***	***Input #2***
ISLD67B017		ISL67B017	ISL67B052
ISLD67B052		ISL67B052	ISL67B072
ISLD67B072	LISLD079	ISL67B072	ISL67B252
ISLD67B112		ISL67B112	ISL67B132
ISLD67B132	LISLD139	ISL67B132	ISL67B312
ISLD67B197	LISLD199	ISL67B197	ISL67B017
ISLD67B252		ISL67B252	ISL67B112
ISLD67B312		ISL67B312	ISL67B197

Br – R+2 toroidal array

PAIR NAME	***former PTDATA name***	***Input #1***	***Input #2***
ISLD79A072		ISL79A072	ISL79A222
ISLD79A147		ISL79A147	ISL79A072
ISLD79A222		ISL79A222	ISL79A272
ISLD79A272		ISL79A272	ISL79A147

Br – R-2 toroidal array

PAIR NAME	***former PTDATA name***	***Input #1***	***Input #2***
ISLD79B067		ISL79B067	ISL79B217
ISLD79B142		ISL79B142	ISL79B067
ISLD79B217		ISL79B217	ISL79B277
ISLD79B277		ISL79B277	ISL79B142

Br – HFS toroidal array (Above midplane)

PAIR NAME	***former PTDATA name***	***Input #1***	***Input #2***
ISLD1A011		ISL1A011	ISL1A199
ISLD1A049		ISL1A049	ISL1A244
ISLD1A109		ISL1A109	ISL1A011
ISLD1A139		ISL1A139	ISL1A341
ISLD1A199		ISL1A199	ISL1A139
ISLD1A244		ISL1A244	ISL1A274
ISLD1A274		ISL1A274	ISL1A109
ISLD1A341		ISL1A341	ISL1A049

Br – HFS toroidal array (Below midplane)

PAIR NAME	*former PTDATA name*	*Input #1*	*Input #2*
ISLD1B011		ISL1B011	ISL1B199
ISLD1B049		ISL1B049	ISL1B244
ISLD1B109		ISL1B109	ISL1B011
ISLD1B139		ISL1B139	ISL1B341
ISLD1B199		ISL1B199	ISL1B139
ISLD1B244		ISL1B244	ISL1B274
ISLD1B274		ISL1B274	ISL1B109
ISLD1B341		ISL1B341	ISL1B049

Br – HFS vertical array

PAIR NAME	*former PTDATA name*	*Input #1*	*Input #2*
ISLD5A199		ISL5A199	ISL5A139
ISLD4A199		ISL4A199	ISL4A139
ISLD3A199		ISL3A199	ISL3A139
ISLD2A199		ISL2A199	ISL2A139
ISLD1A199		ISL1A199	ISL1A139
ISLD1B199		ISL1B199	ISL1B139
ISLD2B199		ISL2B199	ISL2B139
ISLD3B199		ISL3B199	ISL3B139
ISLD4B199		ISL4B199	ISL4B139
ISLD5B199		ISL5B199	ISL5B139

Br – R0 toroidal array (external)

PAIR NAME	*PTDADA Name (if different)*	*Input #1*	*Input #2*
ESLD66M019		ESL66M019	ESL66M079
ESLD66M079	ESLD079U	ESL66M079	ESL66M259
ESLD66M139	ESLD139U	ESL66M139	ESL66M319
ESLD66M199	ESLD199U	ESL66M199	ESL66M019
ESLD66M259		ESL66M259	ESL66M139
ESLD66M319		ESL66M319	ESL66M199

Br – R0 toroidal array (external): with hardware compensation

PAIR NAME	*former PTDATA name*	*Input #1*	*Input #2*
ESLD079		ESL66M079	ESL66M259
ESLD139		ESL66M139	ESL66M319
ESLD199		ESL66M199	ESL66M019

Br – R+1 toroidal array (external)

PAIR NAME	*former PTDATA name*	*Input #1*	*Input #2*
ESLD67A004		ESL67A004	ESL67A244
ESLD67A034		ESL67A034	ESL67A154
ESLD67A064		ESL67A064	ESL67A184
ESLD67A094		ESL67A094	ESL67A274
ESLD67A124		ESL67A124	ESL67A304
ESLD67A154		ESL67A154	ESL67A334
ESLD67A184		ESL67A184	ESL67A004
ESLD67A214		ESL67A214	ESL67A094
ESLD67A244		ESL67A244	ESL67A124
ESLD67A274		ESL67A274	ESL67A034
ESLD67A304		ESL67A304	ESL67A064
ESLD67A334		ESL67A334	ESL67A214

Br – R-1 toroidal array (external)

PAIR NAME	*former PTDATA name*	*Input #1*	*Input #2*
ESLD67B004		ESL67B004	ESL67B244
ESLD67B034		ESL67B034	ESL67B154
ESLD67B064		ESL67B064	ESL67B184
ESLD67B094		ESL67B094	ESL67B274
ESLD67B124		ESL67B124	ESL67B304
ESLD67B154		ESL67B154	ESL67B334
ESLD67B184		ESL67B184	ESL67B004
ESLD67B214		ESL67B214	ESL67B094
ESLD67B244		ESL67B244	ESL67B124
ESLD67B274		ESL67B274	ESL67B034
ESLD67B304		ESL67B304	ESL67B064
ESLD67B334		ESL67B334	ESL67B214

Bt – R0 toroidal array

PAIR NAME	*former PTDATA name*	*Input #1*	*Input #2*
BTID66M053		BTI66M053	BTI66M233
BTID66M132		BTI66M132	BTI66M053
BTID66M233		BTI66M233	BTI66M312
BTID66M312		BTI66M312	BTI66M132

History of Revisions

7/6/16 (ejs)	First version, based on data copied from the excel database.
7/20/17 (ejs)	Note on flux loops added.
1/18/18 (ejs)	Note on loop voltages added. RLIU30, RLC19, … changed to IU30, C19, … (as of early 2017). Added this history of changes at the end of the document.
4/26/18 (sm)	Corrected column headings for NA from m^2 to cm^2 (no change to values). Updated values for MPI2A067 (Z), MPI1U157 (R,Z).
8/6/18 (sm)	Width for ISL79A/B loops (R±2) changed from deg to m, and entered values. HFS saddle loop NA changed from 2426.2 to 2420.2 cm^2, per new calc by ejs.
2/6/19 (ejs)	Merged 1/18/18 and 8/6/18 versions. HFS saddle loop NA returned to 2426.2 cm^2 due to errors in the new calculation. Added notes on naming conventions.
2/11/19 (ejs)	Merged "Magnetic_sensor_coords_2019 v0" (2/6/19) with a temporary version containing many updates to the data tables – see "database updates 20190208".
2/14/19 (ejs)	NA column values and headings changed from cm^2 to m^2.
2/19/19 (ejs)	NA column updated to m^2 for one group of probes that was missed on 2/14.
2/22/19 (ejs)	Added tables of correspondence between MDSplus tags and ptdata pointnames. Merged in the memo "3D Magnetic Arrays_20190212" as Part II of this memo.
5/5/20 (sm)	Bp arrays modified for helicon antenna: 67 and 142 poloidal, R+1 toroidal (pairs) R±1 external Br arrays reinstated. Most column headings of "pdtata name" changed to "former ptdata name"
(ejs)	Updated explanation of ptdata names vs MDSplus tags Changed font to Times New Roman for better formatting, sub/super scripts, etc.